

Original Article

Novel *rrs* mutations in second-line injectable drug-resistant clinical isolates of *Mycobacterium tuberculosis* from the Punjab province of Pakistan

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ABSTRACT

Introduction: Phenotypic drug susceptibility testing is the most common approach to assess drug-resistant isolates; however, molecular methods of drug susceptibility testing are fast, accurate hence, offer less time for transmission during the diagnosis period. As data on the molecular methods regarding injectable drug resistance in the Punjab province of Pakistan is limited, therefore in this study, we aimed to analyze the mutations in the *rrs* gene behind second-line injectable drug resistance.

Material and methods: *Mycobacterium tuberculosis* isolates were collected from the sputum of 5362 TB suspects. The strains confirmed for resistant to injectable drugs through drug susceptibility testing were further proceeded. The 1537bp *rrs* gene was amplified with the help of three sets of primers with overlapping regions and DNA sequencing was performed. Obtained sequences were aligned with reference sequence to find mutations. RFLP-PCR method was also optimized for rapid detection of a common (143bp and 205bp) *rrs* gene mutation.

Results: Among 172 rifampicin resistance isolates, 163(95%) were resistant to both rifampicin and isoniazid, and 9 (5%) were resistant to only rifampicin. Among the resistant samples, 12 (6.9%) samples were resistant to all three injectable drugs. Sixty out of 172 (34.9%) samples showed resistance to at least one drug and 10 (5.8%) samples were resistant to two drugs among the 3 s-line drugs. Sequencing analysis showed novel mutations in different samples at positions 443InsC, 19DelT, 29G>A, 48C>T, 50G>C, 265InsT, 423T>G, 476InsA, 446A>G, 563DelA, 695G>A, 805DelA, 900G>A, and 1510A>G, while some already reported mutations at position 1401A>G, 1402A>G, and 1484G>T were also observed. MIC of novel *rrs* gene mutations in KAN, CAP, and AMK resistant isolates were found between 2.5 mg/L-3.05 mg/L, 2.08 mg/L-3.0 mg/L, and 2.1 mg/L-2.7 mg/L respectively.

Conclusion: Novel mutations in the *rrs* gene reported in this study may confer second-line injectable drugs resistance in Mtb. This molecular insight into second-line injectable drug resistance is useful for better management of resistance Mtb in high burden countries.

1. Introduction

Tuberculosis (TB) remains a main cause of mortality and morbidity worldwide [1]. As an infectious disease, TB is a main reason for bad health. *Mycobacterium tuberculosis* (Mtb) is a rod-shape bacterium that causes the TB disease. In air, Mtb spreads mostly from TB patients when breathing, sneezing, or coughing. There are different types of TB, pulmonary TB (affects lungs) and extra-pulmonary TB (affects other body parts). From the world population, Mtb infects almost a quarter people [2]. Pakistan has a population of more than 200 million and ranks 6th in the list of TB global burden, with a prevalence rate of 359/100,000 and

an occurrence rate of 181/100,000. It has been assessed that 51% of the TB cases are focused in the province of Punjab, followed by 23% in the province of Sindh, 15% in Khyber Pakhtunkhwa and 3.5% in Baluchistan, with rest being divided among Azad Kashmir and northern and tribal areas [3].

TB remains a leading cause of death worldwide due to the emergence of drug resistance. TB is a serious health issue due to the increase in the incidence of multidrug resistance (MDR) rifampicin (RIF) and isoniazid (INH). Certainly RIF and INH are the two most potent TB drugs, and any resistance to these, especially to RIF, may cause the failure of TB therapy. However, TB patients with MDR could be treated using suitable

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combinations of second-line drugs (SLDs). It is commonly believed that drug resistance to TB grows by the selection of drug resistant strains in TB patients who don't complete TB treatment with the suitable combination of medicines or who are having sub-optimal drug regimens. In this condition, it is better to diagnose the level of resistance before it develops into MDR-TB, which can easily spread among populations [4].

For this purpose, drug susceptibility testing (DST) is performed on first-line drugs to find patients with drug-resistant and MDR tuberculosis. Extensively drug-resistant TB (XDR-TB) strains according to the World Health Organization (WHO) and the Centers for Disease Control Prevention (CDC) [5] that XDR-TB as additional drug resistance along with MDR to any fluoroquinolone (FQ) and any injectable anti-TB drugs amikacin (AMK), kanamycin (KAN) or capreomycin (CAP) [6]. WHO has redefined XDR-TB as an infection caused by MTB strain that fulfill the definition of MDR/RR-TB as well as resistant to any fluoroquinolone and at least one additional group A drug (bedaquiline and linezolid). International health agencies, comprising WHO, strongly emphasize on the requirement of information related to drug resistance and XDR from developing countries that have no mechanisms for conducting surveys on drug resistance. Accessible local data mostly from laboratory-based studies based on insignificant sample size, report a growing rate of MDR isolates. However, such research reports require additional investigation [3]. It was reported that the resistance against KAN, AMK and CAP emerges due to mutations in the 16S rRNA (*rrs*) gene at region 1400 (particularly at positions 1401, 1402 and 1484). Previous studies already demonstrated that the A1401G mutation in *rrs* gene can be readily found in 60% KAN resistant isolates and 175% of CAP or AMK resistant isolates. This mutation was suggested as marker of resistance to all three drugs, specifically KAN and AMK [7].

In this study, we detected novel mutations in the *rrs* gene conferring resistance to second-line injectable drugs. The study is useful for the better management of resistance to injectable SLDs in high burden countries.

2. Materials and methods

2.1. Ethical consideration

The present investigation was approved by the Departmental Ethical Review Committee, ref_IMBB/BBBC/21/1223, University of Lahore. Consent was taken from each participant for collecting details regarding disease history as well as socio-demographic profile (age, gender, ethnicity, education level etc.); however, the results were not shared with the patients.

2.2. Study samples

Mtb isolates were collected from the sputum of 5362 TB suspects (2018–2020) from the TB & chest departments and other healthcare centers of Mayo Hospital, Ganga Ram Hospital, Jinnah Hospital, Services Hospital, Shalimar Hospital, Hameed Latif Hospital and Sheikh Zayed Hospital Rahim Yar Khan, as well as those in the private sector.

2.3. Sample culturing and drug susceptibility testing (DST)

Ziehl–Neelsen (ZN) smear microscopy and the GeneXpert (Xpert) test were performed on all MDR-TB suspects. Sputum specimens from patients with RIF-resistance detected (RRD) on Xpert were cultured on Lowenstein–Jensen (LJ) media. After primary isolation, pure cultures were sub-cultured for DST on LJ medium using the standard proportion method. All the MDR samples were sub-cultured for SLDs' testing (KAN, AMK and CAP) and the colonies from samples resistant to SLDs were diluted in 400–600 µl distilled water. The ATCC H37Rv strain of Mtb, known to be sensitive to all drugs, was used as negative control or any isolate giving more than 1% growth on medium containing AMK, KAN and CAP as compared with control will be labeled as resistant strain.

Clinical samples were collected and DST was performed at Pakistan Health Research Council's (PHRC) biosafety level-3 (BSL-3) laboratory, Lahore.

Minimum inhibitory concentrations (MIC) of AMK, KAN, and CAP were determined for those isolates, harboring single and multiple mutations. Drug-containing media and drugs were being stored at 4 °C. Two dilutions of the inoculum as 10² and 10⁴ were being measured and inoculated on the drug-containing media. As a positive control, for each of the specimens, two dilutions were inoculated on a drug-free LJ medium. As a negative control, the known Mtb strain H37RV; susceptible to all drugs; was inoculated and treated. Incubation was carried out at 37 °C. After the incubation, the growth and colony count were observed for control and drug-contained media.

2.4. Primer designing

Primers were designed to amplify the overlapping segments of the 16S rRNA (*rrs*) gene (NC_000962.3), a target of SLDs. Primers sets F1R1, F2R2, and F3R3 were used to amplify the 433bp, 490bp, and 890bp regions of the *rrs* gene respectively. The second set of primers F4R4 was used to amplify the 143bp region of the *rrs* gene (Table 1)

2.5. Genomic DNA extraction, PCR amplification and sequencing

The genomic DNA of KM, AMK and CAP resistance isolates was extracted using the HipurA® Mtb DNA purification kit (Himedia labs) and the concentration was measured using Nano-drop in ng/µl and stored at –20 °C.

PCR assay was performed to identify the mutations associated with drug resistance. The reaction mixture contained 6 µl of MgCl₂, 5 µl dNTPs, 5 µl forward and reverse primers, 1 µl Taq polymerase (Fermentas, Lithuania), 1 µl genomic DNA, and 22 µl of PCR grade water, making up to 50 µl of the final volume after being added. The PCR conditions were as follows: initial denaturation at 95 °C for 3 min, denaturation at 95 °C for 45 s, annealing at 51 °C for 45 sec, extension at 72 °C for 1 min, final extension at 72 °C for 7 min and the number of cycles were 30. The quality of the amplified DNA was checked using 2% agarose gel before genomic sequencing [8].

The reaction mixture for polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) contained forward (F4) and reverse primer (R4) to target specific DNA regions and the amplified product, followed by restriction digestion with Bsh12361 (10U/µg lambda DNA x 16 hours) (Thermo-scientific). Bsh12361 enzyme of 1.5 µl, 2 µl 10x Buffer and 1 µl PCR product, in addition to distilled water make the final volume about 20 µl. After gently mixing and spinning for a few seconds, the reaction mixture was incubated at 37 °C for 1–16 hours. After incubation, the samples were run on 2% agarose gel along with a 50bp DNA ladder. Two fragments of 143bp and 205bp bands were obtained by the PCR-RFLP of the *rrs* gene.

The DNA sequencing of the *rrs* gene was performed for mutation analysis. Sequencing was performed with the help of ABI2120 genetic analyzer automated sequencer (Applied Biosystem, USA).

Table 1
Primer used for Polymerase Chain Reaction.

Primer	Primer Sequence	Amplify DNA region
F1	5'TACCTTTGGCTCCCTTTTCC-3'	433 <i>rrs</i> region
R1	5'CGTCGATGGTGAAAGAGGT-3'	
F2	5'-AGGCAGCAGTGGGGAATA-3'	490 <i>rrs</i> region
R2	5'AAGGAAGGAAACCCACACCT-3'	
F3	5'-GGGAGCGAACAGGATTAGA-3'	844 <i>rrs</i> region
R3	5'-TCGCCCACTACAGACAAGAA-3'	
F4	5'-CGAGGTTAAGCGAATCCTTA-3'	143 <i>rrs</i> core region
R3	5'-TCGCCCACTACAGACAAGAA-3'	

2.6. Data analysis

Sequence data was analyzed on Snap Gene Viewer V.5.0. Data was assessed and compared with the *rrs* (Rvn01) of the Reference Sequence database of NCBI (NC_000962.3).

3. Results

3.1. Pulmonary and extra-pulmonary TB cases and socio-demographic data

A total of 5362 samples were collected from TB suspects (from 2018 to 2020), previously treated for pulmonary and extra-pulmonary TB. The samples were screened through ZN smear microscopy and were subjected to Xpert test on the suspected and previously Mtb treated pulmonary TB patients' samples as well as on extra-pulmonary TB patients' samples, as mentioned in Table 2 and 172/5362 were Mtb/RIF resistant. The socio-demographic data of the TB patients is mentioned in Table 3.

3.2. Sample culturing and DST

A total of 172 Mtb/RIF resistant cases were cultured on LJ Media and incubated for 6–8 weeks; pure cultures were sub-cultured for the DST of first-line Mtb drugs to find that 163 (95%) samples were multidrug-resistant (to RIF and INH) and 9(5%) were only RIF-resistant, as mentioned in Table 2. The DST of SLDs (KAN, AMK and CAP) was also performed and results have been presented in Table 4. The DST results showed that 12 (6.9%) samples were resistant to all three injectable drugs, 60 out of 172 (34.9%) were resistant to at least one of the three injectable drugs, and 10 (5.8%) were resistant to two of the three SLDs tested (Table 4). A total of 82 samples were resistant and 90 were sensitive. Among the resistant 12 samples were resistant against three drugs in which 10 samples showed mutation (Table 5). Two samples did not show any mutation that's could be due to influx or efflux.

MIC of KAN, AMK and CAP were found different for each of the novel mutations on L.J Media. Its values fluctuated in single mutations, but increases in double, triple and quadruple mutations. KAN and CAP have high MIC when compared with AMK as shown in Table 6.

3.3. PCR amplification and sequencing of the *rrs* gene

Out of 163 MDR isolates, 82 were SLD-resistant (Table 4), and they were amplified to identify mutations in the *rrs* gene (16S rRNA)

Table 2
Distribution of pulmonary and extra pulmonary TB cases.

	Classification of cases	New	Previously treated	Total
Pulmonary	Samples examined for Mtb	2741	2620	5361
	TB positive	815	557	1372
	RIF-resistant	87	84	171
	RIF not resistant	722	467	1189
	Indeterminate RIF resistance	06	06	12
	Mtb not detected	1920	2057	3977
	Classification of cases	New	Previously treated	Total
Extra pulmonary	Samples examined for Mtb	210	27	237
	TB Positive	10	5	15
	RIF-resistant	0	1	1
	RIF not resistant	10	4	14
	Indeterminate RIF resistance	0	0	0
	Mtb not detected	200	22	222

Table 3

Demographic characteristics of the study subjects.

Demographic factors	Characteristics	Value N (%)
Gender	Male	74(43)
	Female	97(57)
Marital Status	Married	55 (32)
	Unmarried	103(60)
	Widow/widower	13 (8)
Mean Age in Years	Male	40.87
	Female	37.45
	Total	39.0
Family size	Mean	7.54
	Range	4–13
Education	Illiterate	62 (36)
	Primary	68 (40)
	Middle	34 (20)
	Matric	7 (4)
BCG Scar	Present	144 (84)
	Absent	27 (16)
Family History of TB	Present	106(62)
	Absent	65 (38)
History of contact	Present	120 (70)
	Not established	51 (30)
History of smoking	Present	75 (44)
	Absent	96 (56)
History of anti-tubercular treatment (ATT)	Present	109 (64)
	Absent	62 (36)
Socio economic status	Poor	150 (88)
	Middle Class	21 (12)

Table 4

Resistance pattern of second-line anti-TB drugs.

Number of drugs	Names of drugs	Number of resistant cases	Percentage (%) age)
One drug	KAN	20	11.6%
	CAP	25	14.5%
	AMK	15	8.7%
Two drugs	KAN+CAP	8	4.7%
	KAN+AMK	1	0.6%
	AMA+CAP	1	0.6%
Three drugs	KAN+CAP+AMK	12	7.0%

KAN = Kanamycin; CAP = Capreomycin; and AMK = Amikacin.

including core region. For accurate sequencing, 1600bp of the *rrs* gene was amplified in three segments using primers with overlapping regions (Fig. 1).

The sequencing results showed the *rrs* gene mutations in three regions (Table 5). Seven were found in *rrs* region 1, five were detected in each of *rrs* region 2 and 3 (Table 5 and Fig. 2). Novel mutations in the samples were 443InsC, 19DelT, 29G>A, 48C>T, 50G>C, 265InsT, 423T>G, 476InsA, 446A>G, 563DelA, 695G>A, 805DelA, 900G>A, and 1510A>G. Known mutations were 1401A>G, 1402A>G, and 1484 G>T, which have been already reported.

3.4. PCR-RFLP

The *rrs* gene was screened by PCR and confirmed by PCR-RFLP mutation. PCR-RFLP showed double band at 143bp and 205bp were taken of mutated samples while control band is 348bp, as shown in Fig. 3.

4. Discussion

Molecular methods for the rapid detection of drug resistance are very important to ensure that the patients receive appropriate treatment timely. This study provides molecular insight into *rrs* gene mutations, causing resistance to SLDs in MDR-TB isolates from Punjab, Pakistan. Among anti-TB SLDs, the molecular mechanism of drug resistance to AMK, KAN and CAP is urgently required for better management of TB.

Table 5Mutation regions of the *rrs* gene.

Isolate number	<i>rrs</i> gene region 433	Heterogeneity	<i>rrs</i> gene region 490	Heterogeneity	<i>rrs</i> gene region 844	Heterogeneity
2469	20	Del -/T				
2469	29	G/A				
2469	34	Del -/G				
1205	48	C/T				
1205	50	G/C				
2120	443	Ins C/-				
2292	443	Ins C/-				
808						
2373			265	Ins T/-		
2373			423	T/G		
2373			426	Ins A/-		
2373			446	A/G		
1126						
2156						
2375			695	G/A		
3614					805	Del -/A
2395					900	G/A
2395					1510	A/G
3424					805	Del -/A
2936					563	Del -/A
2427						
1989						

Table 6MIC of novel *rrs* gene mutation of second-line injectable drug resistant.

Isolate number	Mutation (<i>rrs</i> gene)	KAN (mg/L)	CAP (mg/L)	AMK (mg/L)
^a Reference		2.0	2.0	2.0
2120	Insertion C443	2.50	2.08	2.10
2292	Insertion C443	2.35	2.06	2.25
2375	G695A	2.25	2.00	2.15
3614	Deletion 805A	2.60	2.10	2.50
3424	Deletion 805A	2.45	2.08	2.35
2936	Deletion 563A	2.30	2.05	2.18
1205	C48T + G50C	2.55	2.45	2.18
2395	G900A + A1510G	2.75	2.50	2.35
2469	Deletion 20T + G29A + Deletion 34G	2.95	2.65	2.40
2373	Insertion T265 + T423G + Insertion A426 + A446G	3.05	3.00	2.70

^a mg/L: milligram per liter. Reference: H37RV.

Mtb has become resistant against these drugs due to genetic mutations.

In this study, we detected 172 Mtb samples showing resistance to RIF among which 163 (95%) samples were MDR and 9 (5%) was only RIF-resistant. Results of the present study have been found in accordance with a recent study conducted in similar settings, in which 88.1% of INH resistance samples were resistant to RIF on Xpert with a sensitivity of 92.1% [9].

The drug susceptibility testing of SLDs (KAN, AMK, and CAP) was also performed. 12 (6.9%) samples were resistant to all three injectable

drugs, 60 out 172 (34.9%) were resistant to at least one of the three injectable drugs, and 10 (5.8%) were resistant to two of the three SLDs tested presently. A low proportion of 2% XDR-TB were reported among MDR-TB cases in earlier study [10].

Molecular mechanism of drug resistance could be induced by the presence of single gene mutations. In first-line anti-TB drugs, the *rpoB* and *katG* genes are considered most important targets, involved in the resistance against the most powerful anti-TB drugs, RIF and INH [11]. Similarly, some others have also reported that the *gyrA*94 and *rrs*1401 gene mutations may cause XDR-TB among MDR-TB cases [12]. The primer design was based on prior research, which showed that the most common mutation in the world in the *rpoB* gene is located at codon position 531 which constituted 85.67% and codon position 315 for the *katG* gene with around 82% [13]. For SLDs, majority of resistant isolates harbors mutation in *gyrA* gene is located at codon position 94 (79%) [14], and codon position 1401 in *rrs* gene is accounted in 60% [7] resistance isolates. Results of current study are also in agreement with all the afore mentioned studies. AMK, KAN, and CAP are the second most common anti-TB drugs among many drugs [15]. All these antibiotics bind to the 16S rRNA and block protein synthesis. The 1 common binding are leads to the development of cross-resistance to these antibiotics. Resistance to these antibiotics is generally due to mutations in the 16S rRNA gene 1401A>G, 1402C>T, and 1484G>T [16]. Mutations 1401, 1402, and 1484 in the *rrs* gene in Mtb is usually associated with resistance to AMK and KAN. The most common mutation resistance is 1401A>G. Of the 29 continuous isolates, 1401A>G mutation was found in 21 [17]. Yuan et al. (2014) observed the mutation 1401 A>G as the most common mutation in strains resistant to KAN, AMK and CAP [18].

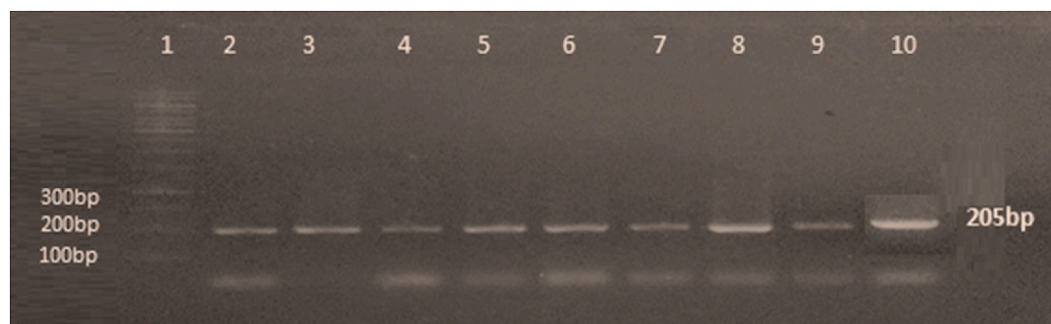


Fig. 1. Lane 1 shows DNA ladder; Lane 2–10 represent PCR amplified products, with temperature ranging from 51 to 55 °C.

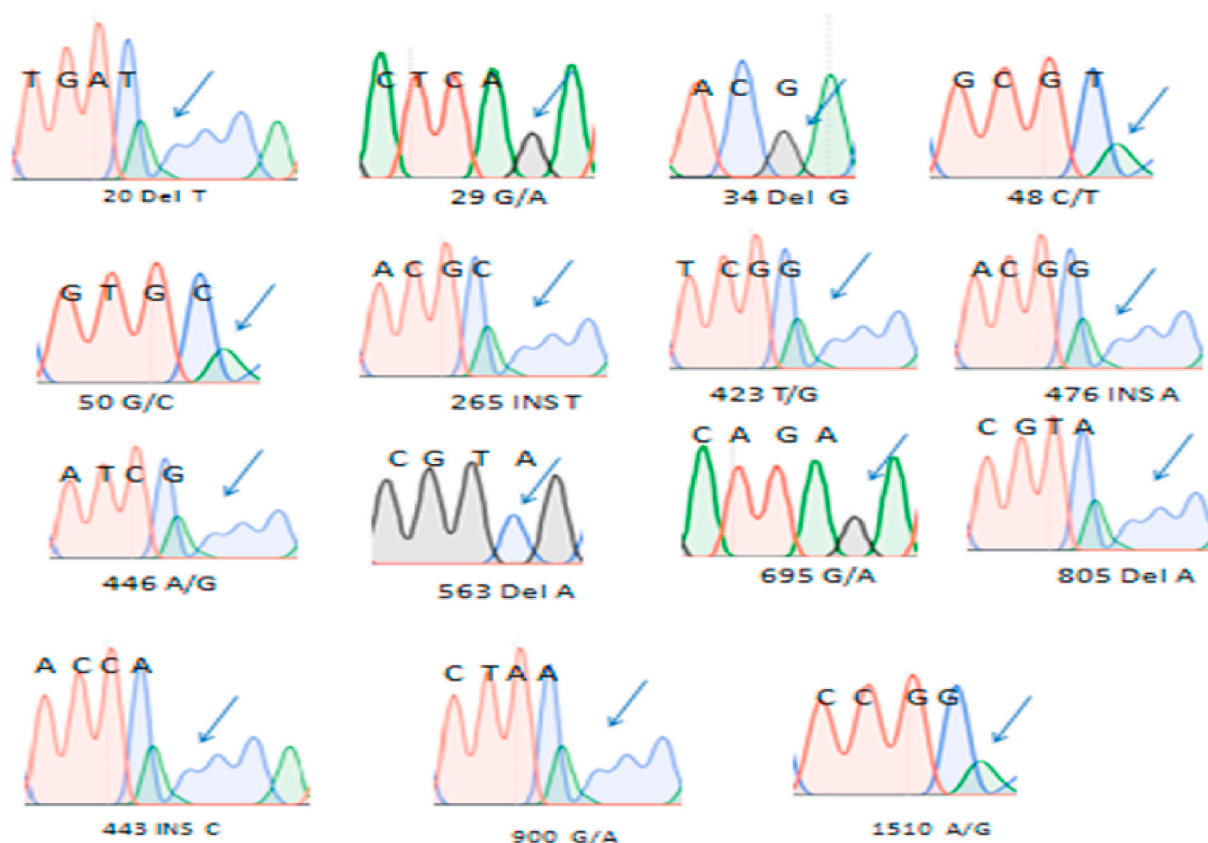


Fig. 2. Sequencing analysis: arrows indicated the mutation.

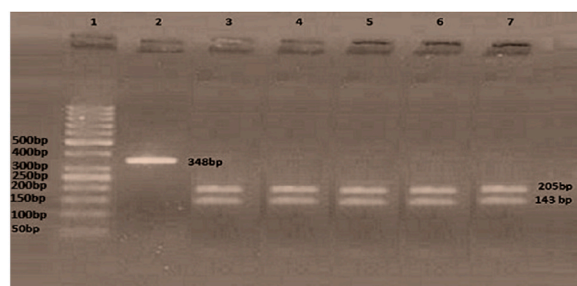


Fig. 3. PCR-RFLP gel electrophoresis. Lane 1 DNA 50bp ladder; lane 2 shows 348bp undigested DNA fragment; lane 3 to 7 show 143bp and 205bp DNA fragments after enzyme digestion.

The 1401 A>G and 1484 G>T mutations in the *rrs* gene were reduced and the 1401 A>G, 1402 C>T and 1484 G>T mutations were observed in the *rrs* gene, the 1484 G>T mutation was detected in only one sample [19]. A study from India reported that mutations in the *rrs* genes 1401A>G and 1884 G>T occurred in clinical *Mtb* resistant to a wide range of drugs. Previous studies have also been reported 1402C>T and 1484G>T mutations in resistant strains. Due to its high correlation with resistance, the 1401 A>G mutation is generally used as a marker for the diagnosis of AMK and KAN resistance. Other mutations are less associated with resistance and are therefore not used as markers [7]. The 1484 G>T mutation has been reported to be associated with resistance to AMK, KAN, and CAP [20], and a high prevalence of the 1484 G>T mutation (17.2%) [21]. The 1484 G>T mutation has previously been observed in *Mtb* strains from India [17].

First antibiotic drug used for the therapeutic control of TB was streptomycin, but due to their side effects and resistance, it was

eliminated from the 1st line anti TB drug group. However, in confirmed streptomycin susceptible strains, WHO recommend to continue the use of streptomycin as it inhibits the protein synthesis of slow growing bacteria. Streptomycin showed irreversible binding to S12 protein and 16S rRNA, the small subunit of bacterial ribosome [22].

Mutations at positions 1401, 1402 and 1484 have been already reported and MIC of KAN, AMK and CAP were found as >40, >20 and 5–15 mg/L [23].

Kardan (2020) reported that the MICs of second-line drugs (KAN, AMK, and CAP) may change due to mutations, and found that among 9 out of 35 samples showed resistance for which MICs were in different range. In the current study mutant isolates exhibited different MIC in a range 2–4 mg/L. Isolates with single mutations have little change in MIC as compared to double and triple mutations. The effect on the MIC of single, double and triple mutant isolates (2936, 1205, 2395, 2469) gradually increases. These results show that infected persons with such isolates may require high doses of drugs for better treatment outcomes [24]. Dijkstra (2017) reported that (84%) of the strains are sensitive to AMK (MIC 2 mg/L) and (9%) to KAN and (7%) to CAP. These results indicate that AMK has high activity and may have low MIC. In our study, MIC of KAN seems high when compared with CAP and AMK [25].

To increase the sensitivity and specificity of molecular diagnostics for the detection of XDR-TB, it is necessary to understand the frequency and geographical distribution of various mutations in the target area of the drug.

5. Conclusion

This study characterizes molecular basis of resistant to second-line drugs in *Mtb* isolates from Punjab, Pakistan. Novel mutations were detected in *rrs* gene at positions 443InsC, 19DelT, 29G>A, 48C>T, 50G>C, 265InsT, 423T>G, 476InsA, 446A>G, 563DelA, 695G>A, 805DelA, 900G>A, and 1510A>G. These findings provide valuable

information for the development of new, rapid, and reliable molecular approaches for the detection of resistance in clinical isolates of Mtb.

ICMJE statement

S. Khurshid and MI Sarwer contributed to the conception and design of the study and experimental work. MI Sarwer contributed to acquisition of data, S. Khurshid and MT Khan did analysis and interpretation of data. All authors contributed to the writing of the final manuscript and final approval of the version to be submitted.

Declaration of competing interest

No author has any type of conflict of interest.

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